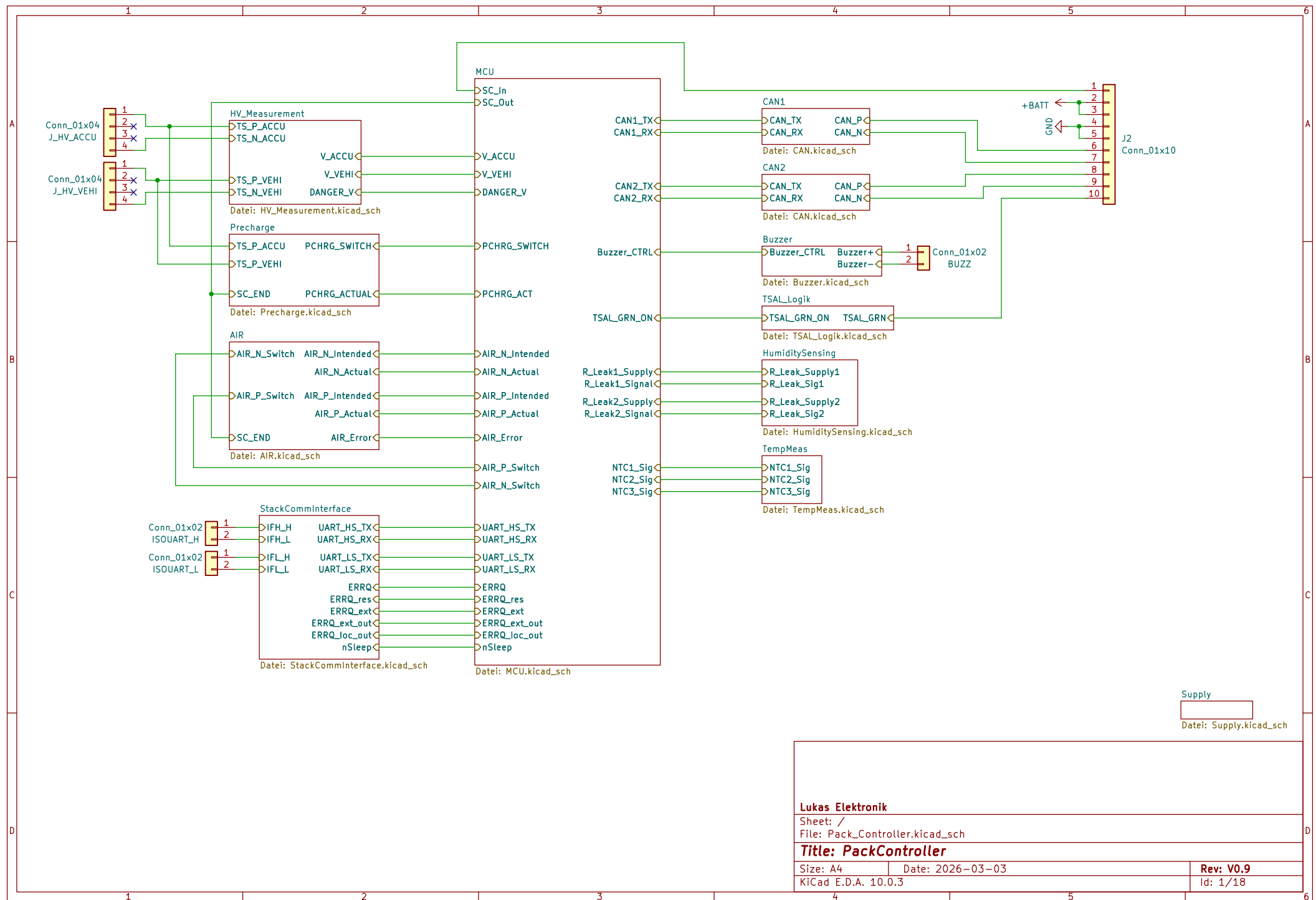
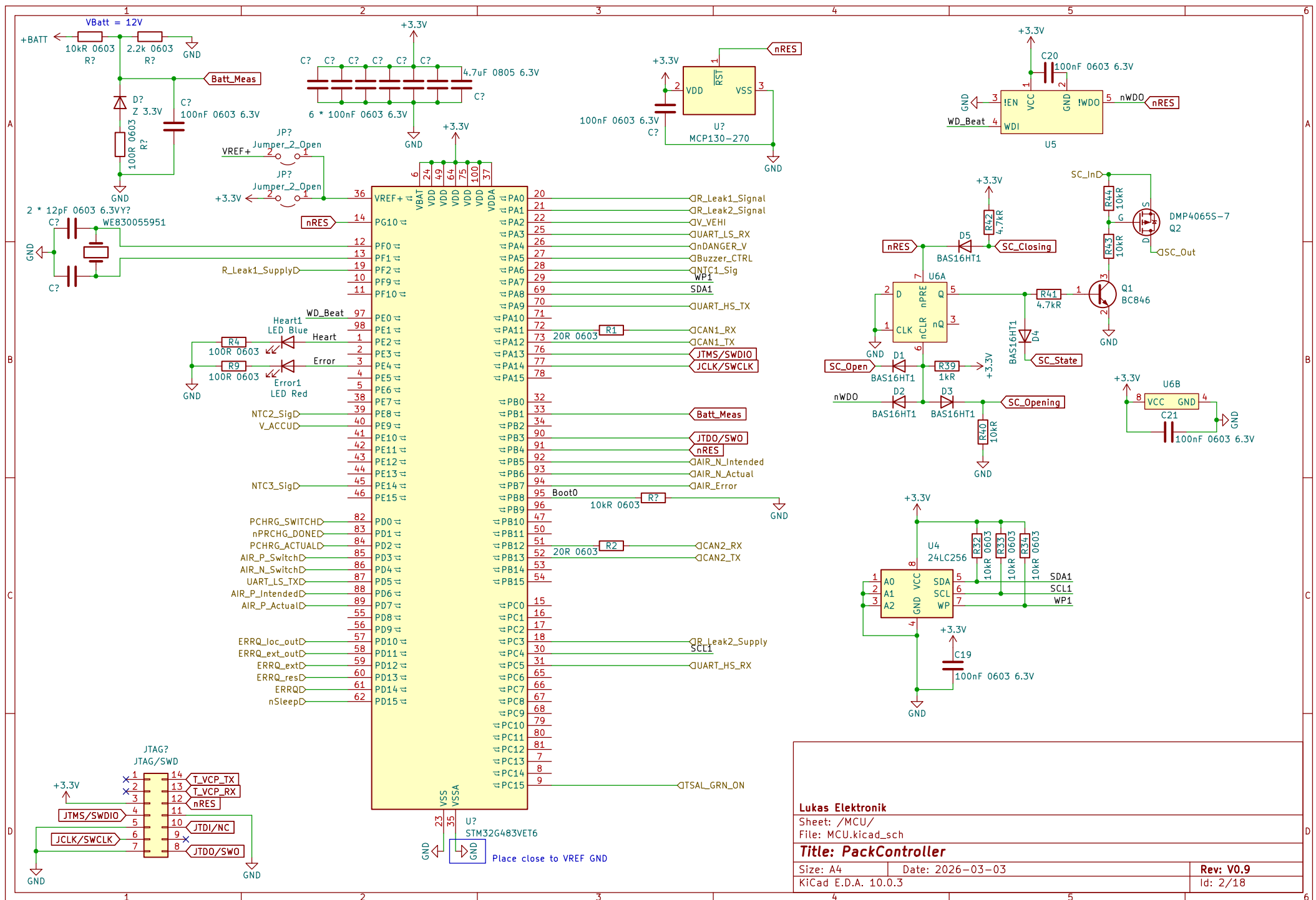






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Sheet: /MCU/

File: MCU.kicad\_sch

**Title:** PackController

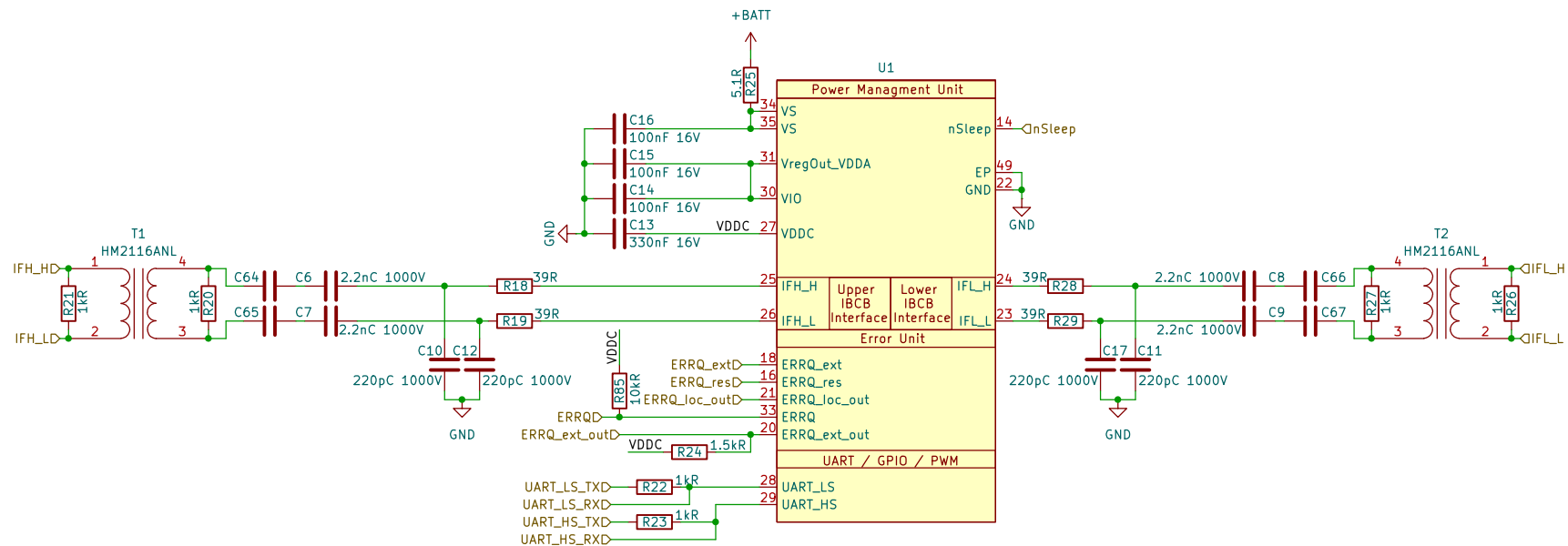
Size: A4

Date: 2026-03-03

|                     |   |
|---------------------|---|
| Size: A1            | 1 |
| KiCad E.D.A. 10.0.3 |   |

Rev: V0.9

Id: 2/18



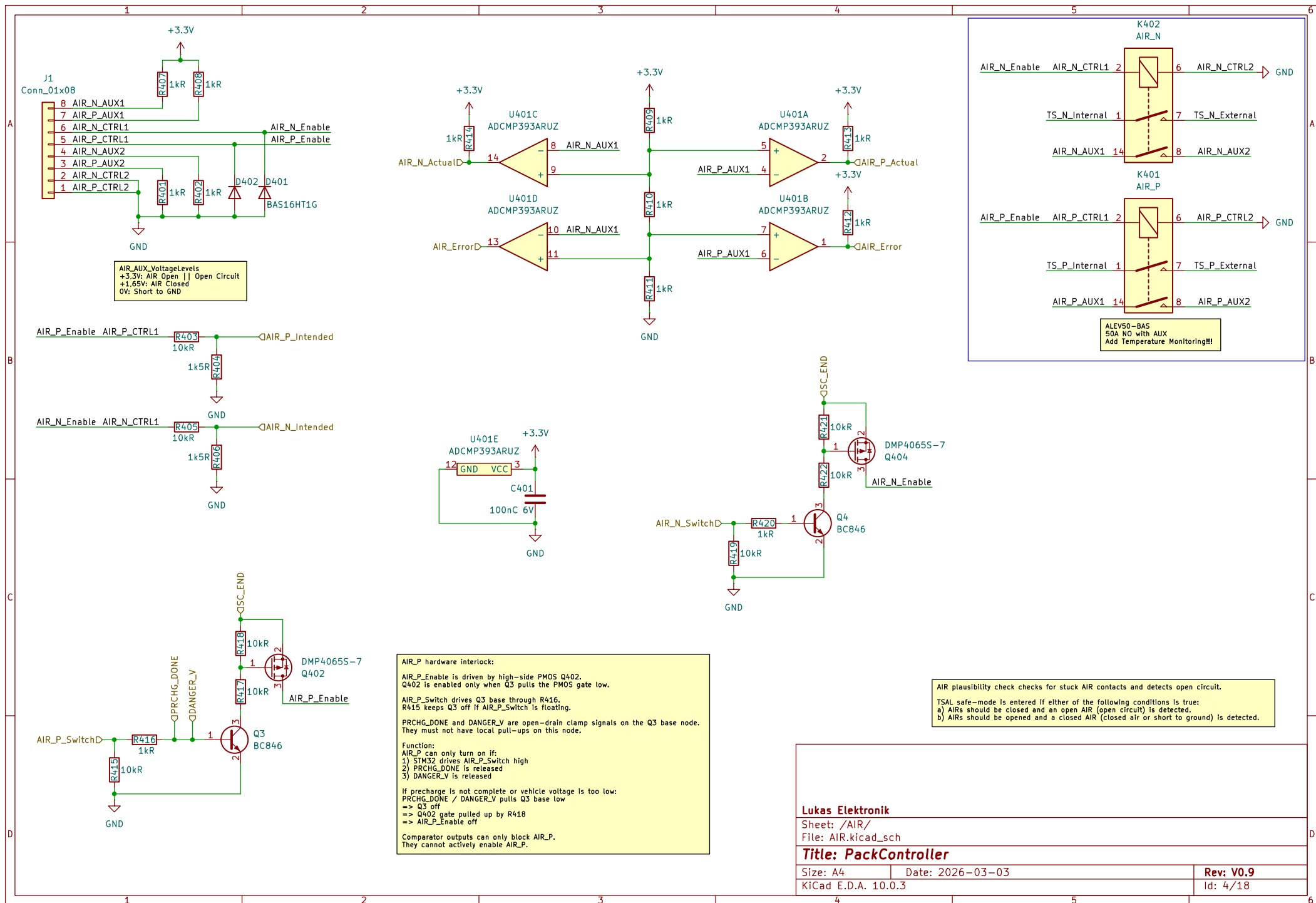
Lukas Elektronik

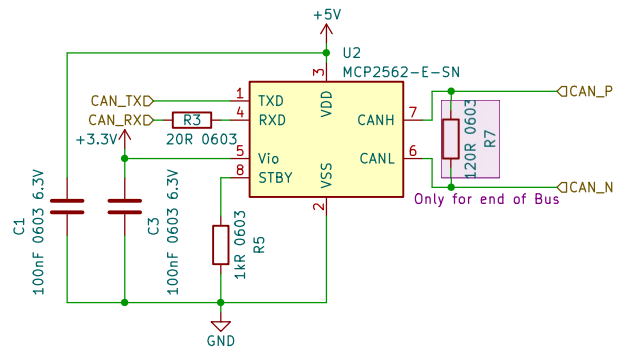
Sheet: /StackCommInterface/  
File: StackCommInterface.kicad\_sch

**Title: PackController**

Size: A4 Date: 2026-03-03  
KiCad E.D.A. 10.0.3

Rev: V0.9  
Id: 3/18





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Sheet: /CAN1/  
File: CAN.kicad\_sch

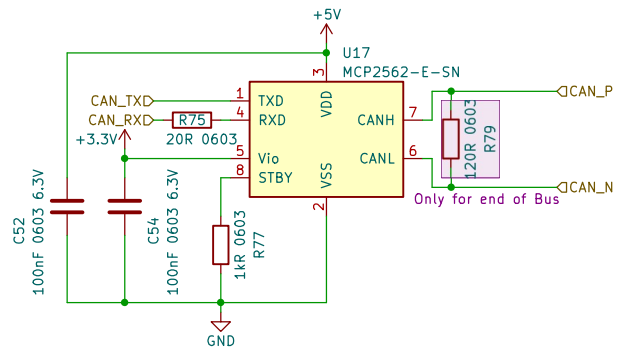
**Title: PackController**

Size: A4 Date: 2026-03-03

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**Rev: V0.9**

Id: 5/18



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Sheet: /CAN2/  
File: CAN.kicad\_sch

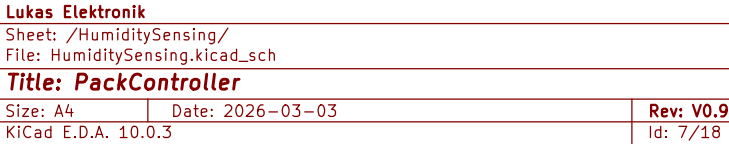
**Title: PackController**

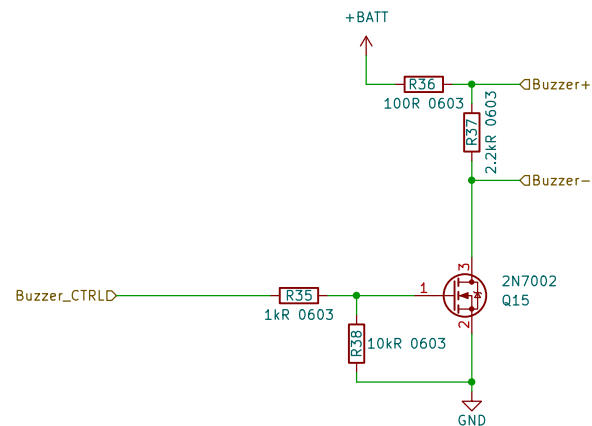
Size: A4 Date: 2026-03-03

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**Rev: V0.9**

Id: 6/18





Passive piezo buzzer driver.  
 Target part: CPT-2305-90PM, externally driven, 12 Vp-p, 4 kHz, typ. 24 nF.

Drive with STM32 PWM:  
 $f_{PWM} \approx 4 \text{ kHz}$   
 $duty \approx 50 \%$

Q15 is a low-side NMOS switch.  
 R37 discharges the piezo during MOSFET off-time:  
 $\tau = R37 * C_{PIEZO} \approx 2.2k * 24nF \approx 53\mu s$ .

R36 limits peak current and EMI.  
 No additional series coupling capacitor required.

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Sheet: /Buzzer/

File: Buzzer.kicad\_sch

**Title: PackController**

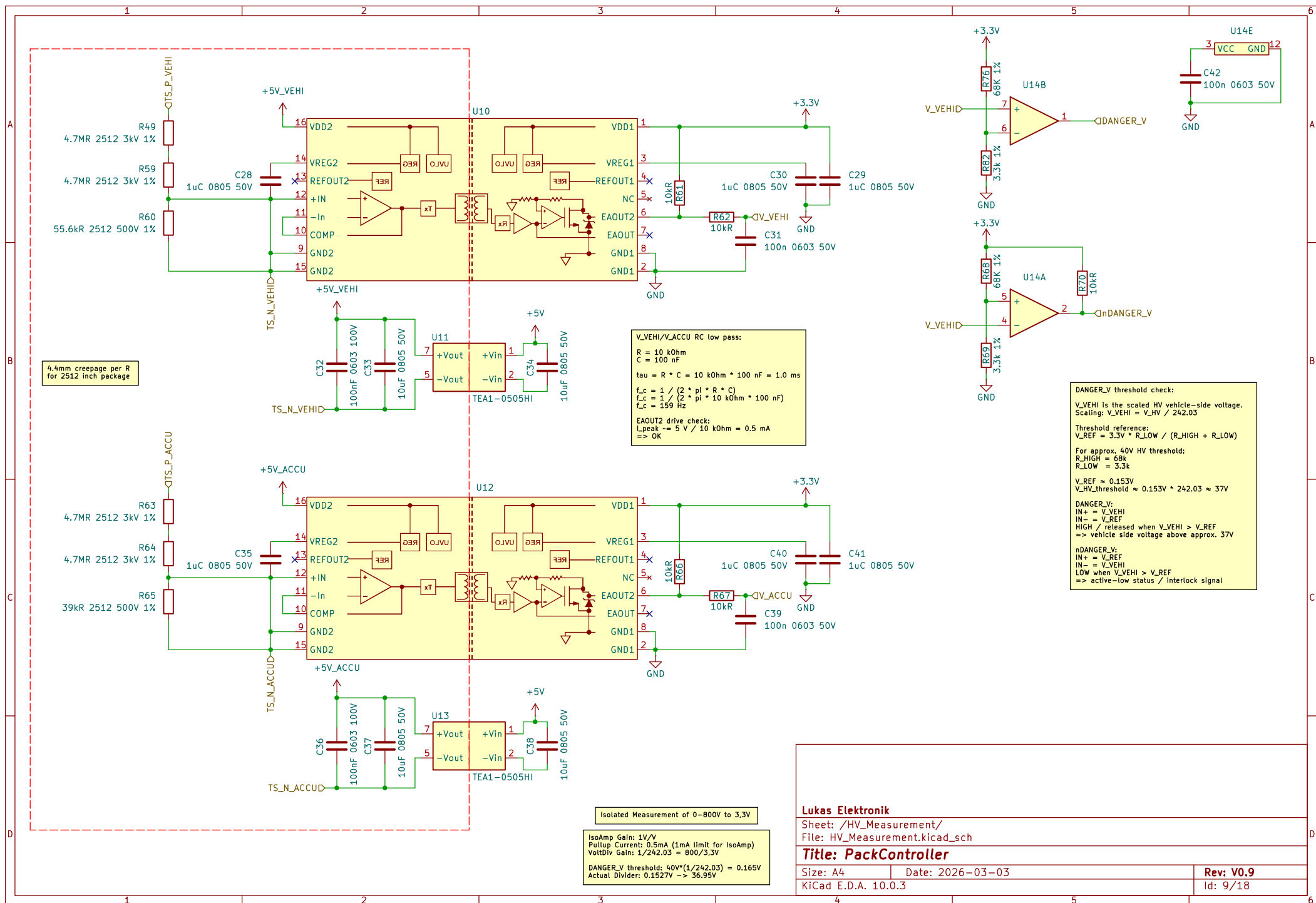
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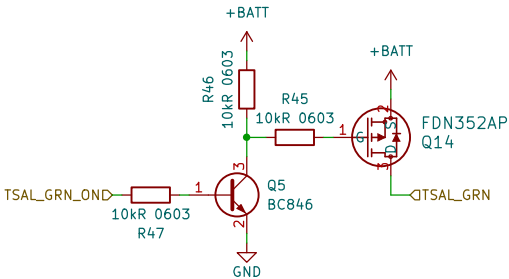
Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 8/18





|                                                   |                  |           |
|---------------------------------------------------|------------------|-----------|
| Lukas Elektronik                                  |                  |           |
| Sheet: /TSAL_Logik/<br>File: TSAL_Logik.kicad_sch |                  |           |
| Title: PackController                             |                  |           |
| Size: A4                                          | Date: 2026-03-03 | Rev: V0.9 |
| KiCad E.D.A. 10.0.3                               |                  | Id: 10/18 |

Peak Strom bei 800V limitiert durch PTC's R25 = 1kOhm für 2S. -> 0,8A  
Toleranz PTC<sub>EL</sub> = 30% -> 0,8/0,7 = 1,14A  
Q7 (IXTA08N100D2HV) Max Cont = 1,8A @ 100°C T<sub>Die</sub>, Pulsed peak = 5A

Zwei Dioden in Reihe für  
Isolation zwischen Fahrzeug  
und HV-Batt  
3\*V<sub>MAX\_HVBatt</sub> -> 3\*800 =  
2400V  
V<sub>RSM</sub> 4kV  
1,7V V<sub>f</sub> @100mA

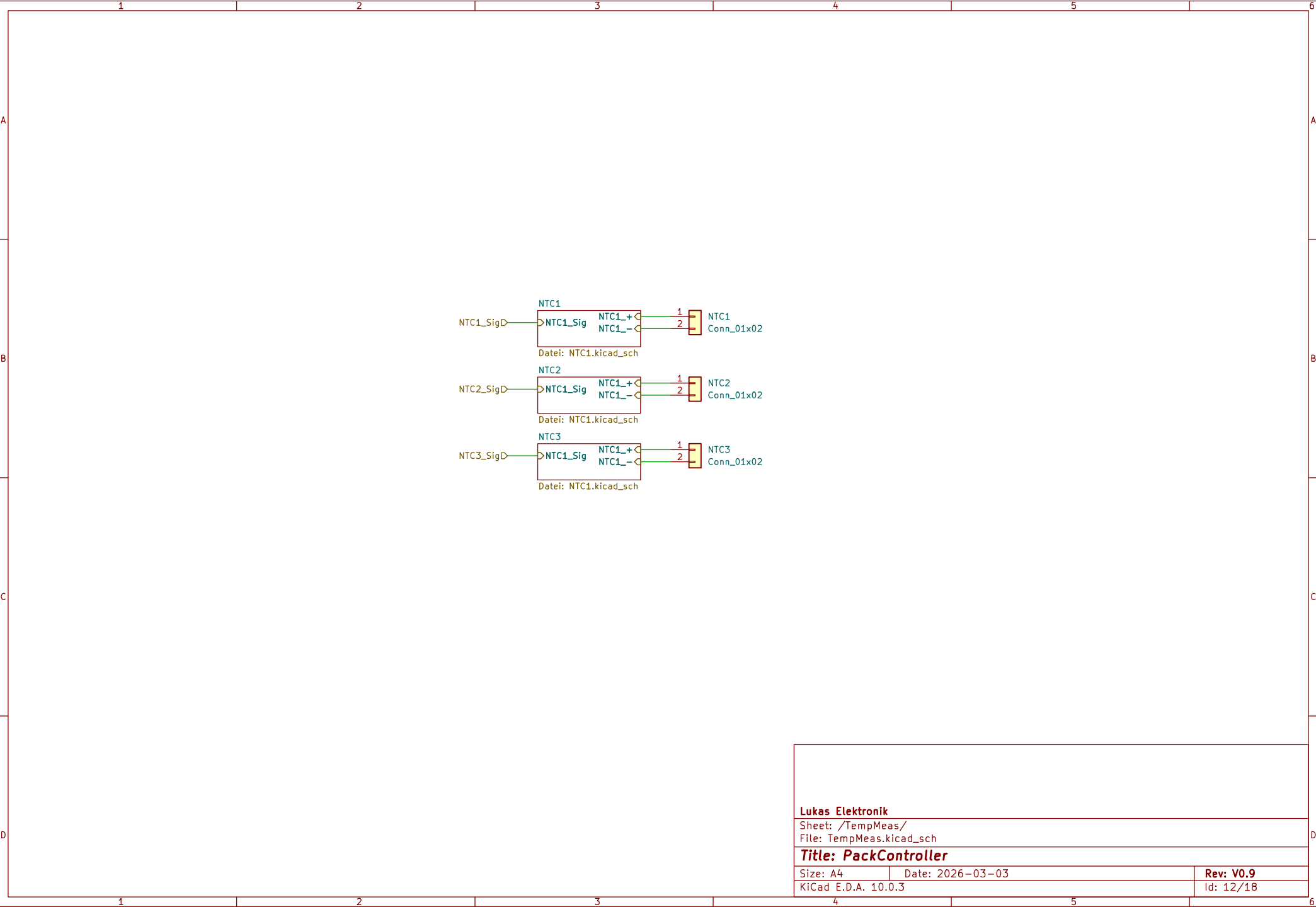
Precharge logic state  
PC open: PCHRG\_ACT = GND  
PC closed: PCHRG\_ACT = 3.3V

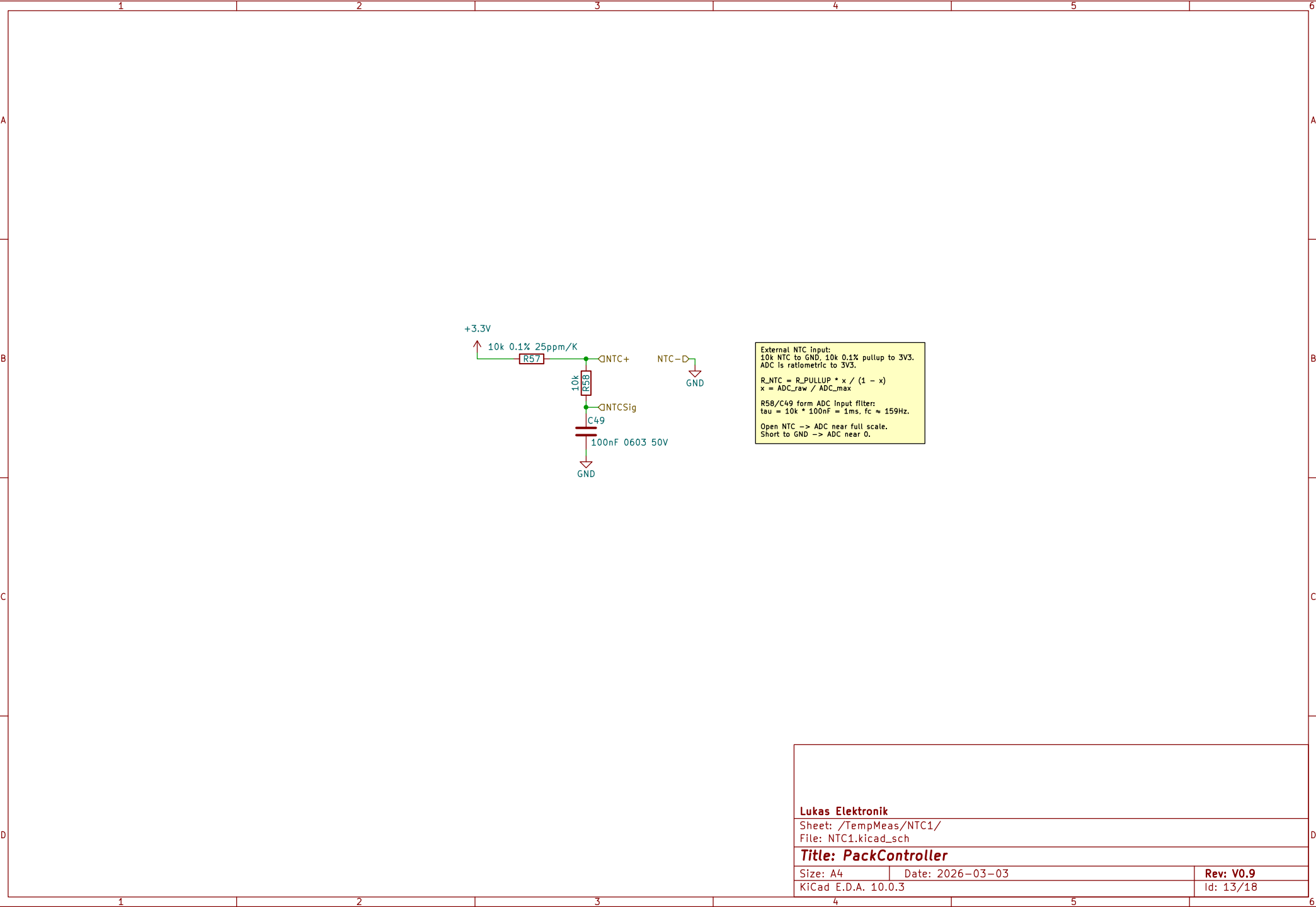
Haltezeit Relais K1  
R = 800Ohm  
C = 47uF (20%) min = 37,6uF  
tau = 3,008ms  
Vt = 0,55V (Drop Out KTL12)  
Vi = 5V  
 $\tau = -R \cdot C \cdot \ln(V_t/V_i)$   
tau = -3,008ms \* ln(0,55V/5V)  
tau = -3,008ms \* -2,207  
tau = 6,64ms  
Q6\_t<sub>off</sub> = 4,8ms

Q6 PRECHARGE GATE TIMING / WORST CASE  
Bestückung:  
R50 = 10 kOhm  
R51 = 10 kOhm  
R52 = 1 kOhm  
R53 = 10 kOhm  
R54 = 1 kOhm  
C22 = 200 nF  
U7, U8 = TC1T1004  
D6 + D7 = ca. 1.7 V  
Annahmen:  
+5 V liegt vor D6/D7  
V<sub>C,U7</sub> = 5.0 V - 1.7 V = 3.3 V  
V<sub>CEsat,U7</sub> max = 0.3 V  
=> V<sub>G,Q6,final,worst</sub> = 3.3 V - 0.3 V = 3.0 V  
U7 Eingang:  
IF<sub>U7,min</sub> = (10 V - 1.6 V) / 1 kOhm = 8.4 mA  
IF<sub>U7,max</sub> = (16 V - 1.25 V) / 1 kOhm = 14.8 mA  
IF<sub>limit</sub> = 60 mA  
=> U7 LED Strom OK  
R54 Verlustleistung:  
P<sub>R54</sub> -> ((16 V - 1.25 V)<sup>2</sup> / 1 kOhm) = 0.218 W  
=> R54 >= 0.25 W, Derating beachten  
U7 Ausgangsreserve:  
IC<sub>U7,min</sub> = CTR<sub>min</sub> \* IF<sub>U7,min</sub>  
IC<sub>U7,min</sub> = 1.60 \* 8.4 mA = 13.4 mA  
U7 Strombedarf bei V<sub>G</sub> = 3.0 V:  
L<sub>R51</sub> = 3.0 V / 10 kOhm = 0.30 mA  
L<sub>R52,U8</sub> = (3.0 V - 1.6 V) / 1 kOhm = 1.40 mA  
L<sub>C22\_start</sub> = 3.0 V / 10 kOhm = 0.30 mA  
L<sub>U7\_req,max</sub> = 2.0 mA  
=> U7 Reserve: 13.4 mA / 2.0 mA = Faktor 6.7  
=> Gate-Endspannung Q6 worst case ca. 3.0 V

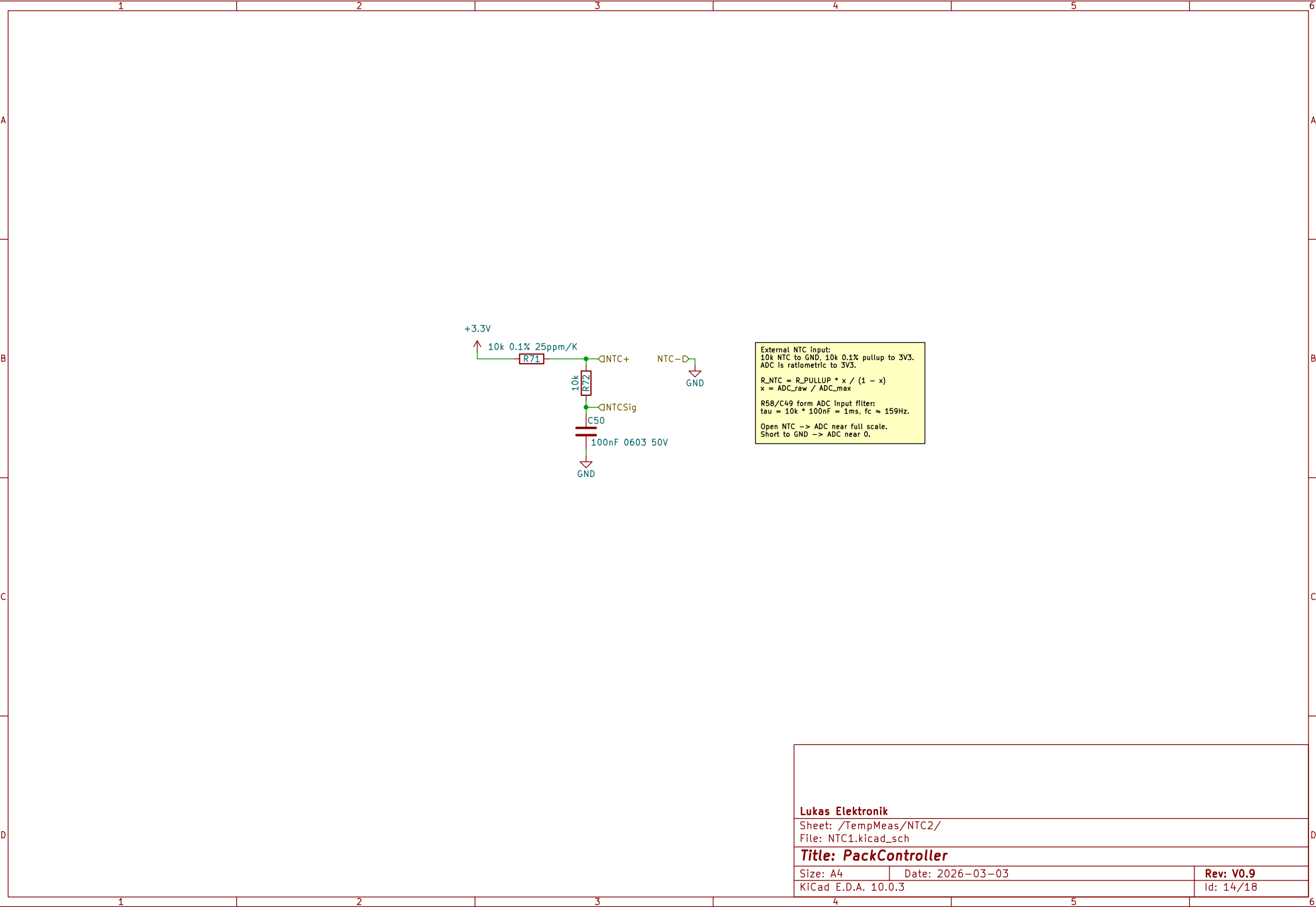
Q6 Einschalten:  
tau<sub>ON</sub> = R50 \* C22  
tau<sub>ON</sub> = 10 kOhm \* 200 nF = 2.0 ms  
Q6 Schwellen:  
VGS(th) = 1.0...2.5 V  
Worst case fuer sicher leitend: VGS(th,max) = 2.5 V  
t<sub>ON</sub> = -tau<sub>ON</sub> \* ln(1 - VGS(th,max) / V<sub>G,final</sub>)  
t<sub>ON</sub> = -2.0 ms \* ln(1 - 2.5 V / 3.0 V)  
t<sub>ON</sub> = 3.6 ms  
=> Q6 erreicht VGS(th,max) nach ca. 3.6 ms  
Q6 Ausschalten:  
Konservativ: Entladung nur ueber R50 + R51  
R52/U8-Pfad ignoriert, daher maximale OFF-Zeit.  
tau<sub>OFF</sub> = (R50 + R51) \* C22  
tau<sub>OFF</sub> = (10 kOhm + 10 kOhm) \* 200 nF = 4.0 ms  
Worst case Startpegel fuer OFF:  
V<sub>G,start,max</sub> = 3.3 V  
Sicher nicht leitend unter VGS(th,min) = 1.0 V:  
t<sub>OFF</sub> = tau<sub>OFF</sub> \* ln(V<sub>G,start,max</sub> / VGS(th,min))  
t<sub>OFF</sub> = 4.0 ms \* ln(3.3 V / 1.0 V)  
t<sub>OFF</sub> = 4.8 ms  
=> Q6 sicher unter VGS(th,min) nach ca. 4.8 ms  
U8 / PCHRG\_ACT:  
IF<sub>U8,min</sub> = (3.0 V - 1.6 V) / 1 kOhm = 1.4 mA  
CTR<sub>U8,min</sub> = 56 % bei IF = 1 mA  
IC<sub>U8,min</sub> = 0.56 \* 1.4 mA = 0.78 mA  
Fuer PCHRG\_ACT -> 3.0 V an R53:  
L<sub>R53</sub> = 3.0 V / 10 kOhm = 0.30 mA  
=> U8 Reserve: 0.78 mA / 0.30 mA = Faktor 2.6  
=> PCHRG\_ACT wird worst case sauber HIGH  
Ergebnis:  
Q6 ON sicher nach ca. 3.6 ms  
Q6 OFF sicher nach ca. 4.8 ms  
U7, U8 CTR-Reserve ausreichend

Precharge 90% check:  
V<sub>ACC</sub> and V<sub>VEH</sub> are equally scaled HV measurements.  
V<sub>ACC\_90</sub> = V<sub>ACC</sub> \* R<sub>LOW</sub> / (R<sub>HIGH</sub> + R<sub>LOW</sub>)  
R<sub>HIGH</sub> = 1.00M  
R<sub>LOW</sub> = 9.09M  
V<sub>ACC\_90</sub> ≈ 0.900 \* V<sub>ACC</sub>  
PRCHG\_DONE:  
IN+ = V<sub>VEH</sub>  
IN- = V<sub>ACC\_90</sub>  
HIGH when V<sub>VEH</sub> >= 0.9 \* V<sub>ACC</sub>  
Used as MCU status input.  
AIR\_P Interlock:  
Same comparator condition is also used as open-drain clamp without pullup at AIR\_P driver.  
Comparator can only block AIR\_P, not enable it.  
Firmware should close AIR\_P only after V<sub>VEH</sub> >= 95% \* V<sub>ACC</sub> with debounce.  
Hardware lockout threshold: 90%

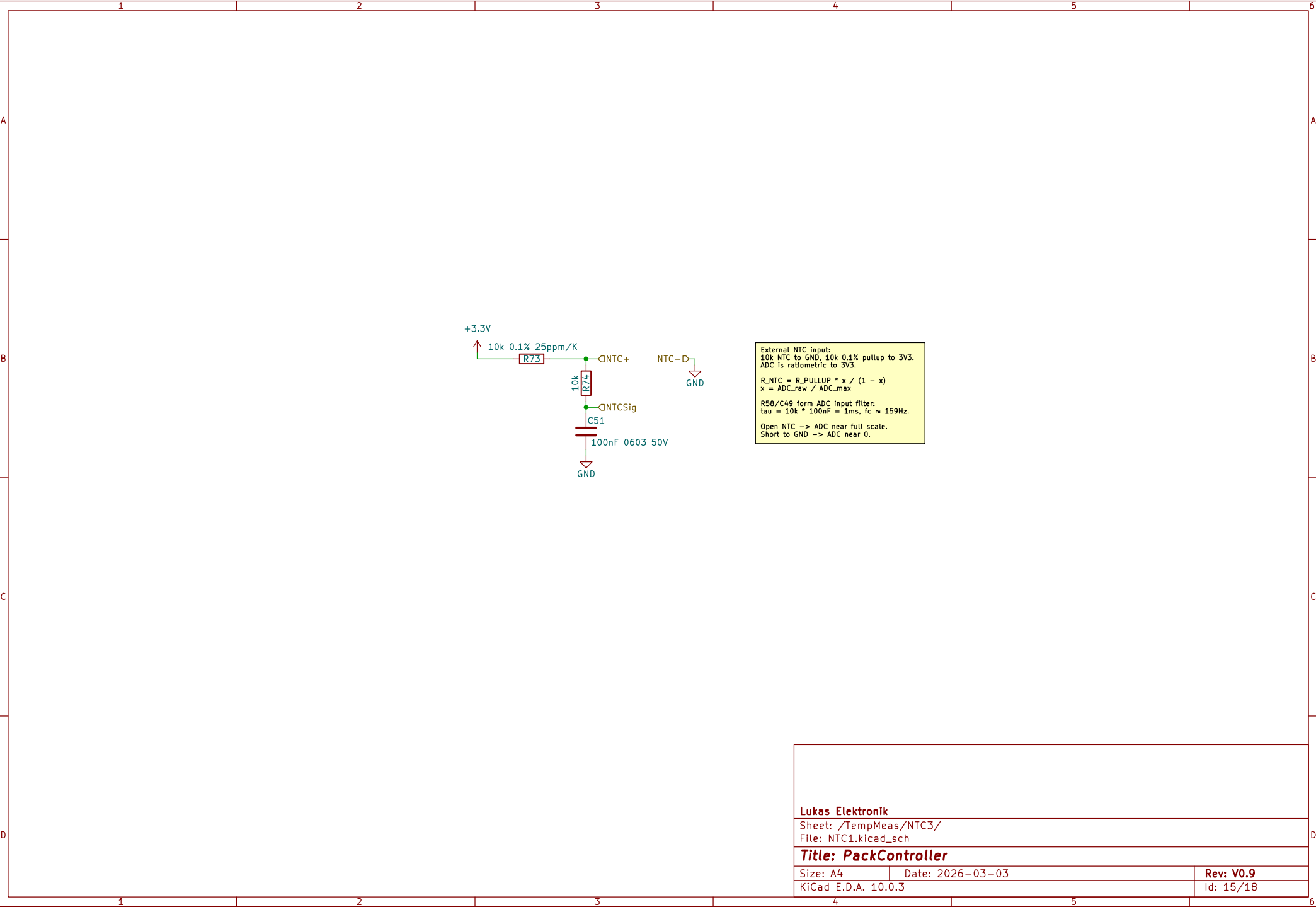




|                        |                  |           |
|------------------------|------------------|-----------|
| Lukas Elektronik       |                  |           |
| Sheet: /TempMeas/NTC1/ |                  |           |
| File: NTC1.kicad_sch   |                  |           |
| Title: PackController  |                  |           |
| Size: A4               | Date: 2026-03-03 | Rev: V0.9 |
| KiCad E.D.A. 10.0.3    |                  | Id: 13/18 |



|                        |                  |           |
|------------------------|------------------|-----------|
| Lukas Elektronik       |                  |           |
| Sheet: /TempMeas/NTC2/ |                  |           |
| File: NTC1.kicad_sch   |                  |           |
| Title: PackController  |                  |           |
| Size: A4               | Date: 2026-03-03 | Rev: V0.9 |
| KiCad E.D.A. 10.0.3    |                  | Id: 14/18 |



Leakage sensor:  
Flex electrodes form R\_LEAK. R\_EOL = 10M is mounted on the flex sensor,  
parallel to the electrodes, for open-wire detection.

$R\_SENSOR = R\_LEAK || R\_EOL$

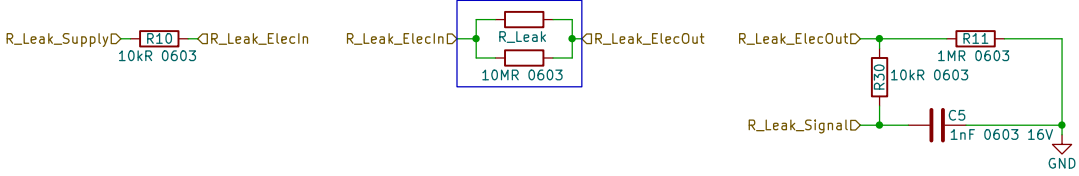
Measurement:  
Pulse R\_Leak\_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/HI-Z.  
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R\_Leak\_Supply = VDD:  
 $x = ADC\_raw / ADC\_max$   
 $R\_SENSOR = R\_REF * (1/x - 1) - R\_SER\_SUPPLY$

Nominal:  
R\_SER\_SUPPLY = 10k  
R\_REF = 1M  
R\_ADC\_SER = 10k  
C\_ADC = 1nF  
R\_EOL = 10M on flex

Expected:  
disconnected/open: ADC ≈ 0  
dry connected: R\_SENSOR ≈ 10M  
wet/leakage: R\_SENSOR decreases

Initial thresholds, validate with real coolant:  
warning: R\_SENSOR < 3M  
leak: R\_SENSOR < 1M  
severe leak: R\_SENSOR < 300k



Leakage sensor:  
Flex electrodes form R\_LEAK. R\_EOL = 10M is mounted on the flex sensor,  
parallel to the electrodes, for open-wire detection.  
 $R\_SENSOR = R\_LEAK || R\_EOL$

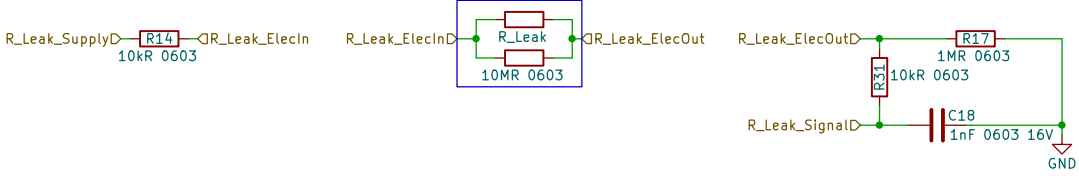
Measurement:  
Pulse R\_Leak\_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/HI-Z.  
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R\_Leak\_Supply = VDD:  
 $x = ADC\_raw / ADC\_max$   
 $R\_SENSOR = R\_REF * (1/x - 1) - R\_SER\_SUPPLY$

Nominal:  
R\_SER\_SUPPLY = 10k  
R\_REF = 1M  
R\_ADC\_SER = 10k  
C\_ADC = 1nF  
R\_EOL = 10M on flex

Expected:  
disconnected/open: ADC ≈ 0  
dry connected: R\_SENSOR ≈ 10M  
wet/leakage: R\_SENSOR decreases

Initial thresholds, validate with real coolant:  
warning: R\_SENSOR < 3M  
leak: R\_SENSOR < 1M  
severe leak: R\_SENSOR < 300k

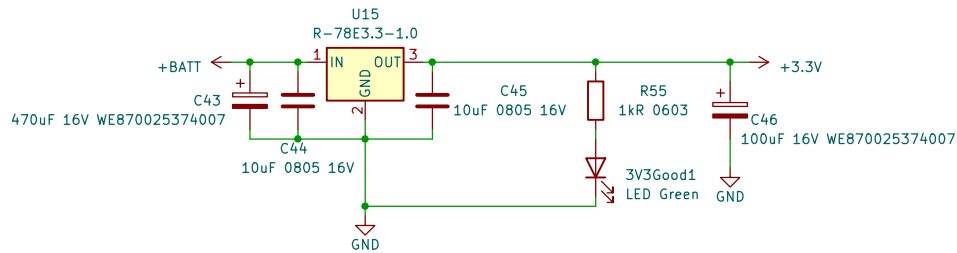


**+3V3 Supply worst-case estimate:**

Main loads:  
STM32G483 active incl. peripherals: -50 mA  
ADuM3190 VDD1, 2x: -4 mA  
24LC256 EEPROM write: -3 mA  
MCP2562 VIO, 2x worst-case: -1 mA  
ADCMP393 comparators + logic: <1 mA  
Pullups, dividers, GPIO drivers, LEDs: -20...30 mA

Estimated worst-case load: -80...90 mA  
Selected supply: RECOM 3.3 V / 1 A DC/DC

Design margin:  
1 A / 90 mA ≈ 11x

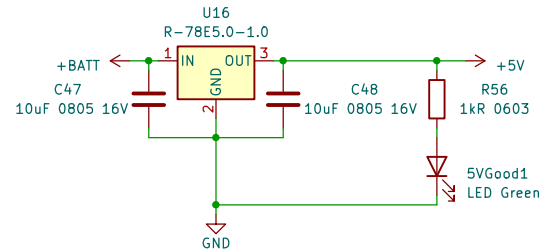


**+5V Supply worst-case estimate:**

Main loads:  
TEA1-0505HI Isolated DC/DC no-load, 3x: -84 mA  
TEA1 loads incl. ADuM3190 Isolated sides: -25...30 mA  
MCP2562 CAN transceivers, 2x dominant: -140 mA  
Misc. reserve / startup / tolerance: -50 mA

Estimated worst-case load: -250...300 mA  
Selected supply: RECOM 5 V / 1 A DC/DC

Design margin:  
1 A / 300 mA ≈ 3.3x



**Lukas Elektronik**

Sheet: /Supply/  
File: Supply.kicad\_sch

**Title: PackController**

Size: A4 Date: 2026-03-03  
KiCad E.D.A. 10.0.3

**Rev: V0.9**  
Id: 18/18